Sensitivity of Turbulent Lifted Hydrogen Jet Flame

to Bulk Velocity and Ignition Location

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Master of Philosophy

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August 2026

To my parents — Mr and Mrs Adewoye.
for their love, support, and sacrifices through the years.

Declaration

This thesis is the result of my own work and includes nothing which is the outcome of work done in collaboration or content generated by artificial intelligence, except as declared in the preface and specified in the text. It is not substantially the same as any work that has already been submitted, or is being concurrently submitted, for any degree, diploma or other qualification at the University of Cambridge or any other University or similar institution except as declared in the preface and specified in the text. It does not exceed the prescribed word limit for the relevant Degree Committee.

Samson Oluwatosin Adewoye
August 2026

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During this MPhil journey, there have been many people who held me through the rough roads and without their help, I would never have made it to the finish line and thereby writing this thesis.

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Finally, I am deeply grateful to my family for their unconditional love and constant support. I could not have achieved this without them. I would like to give special thanks to Mr. and Mrs. Aniemeka, and Miss Yvette Thompson for their regular calls and long conversations, which helped me stay in a good and positive state of mind.

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Abstract

This is where you write your abstract ...

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Nomenclature

Roman Symbols

F complex function

Greek Symbols

γ a simply closed curve on a complex plane

$\imath$ unit imaginary number $\sqrt{-1}$

π $\simeq 3.14\ldots$

Superscripts

j superscript index

Subscripts

0 subscript index

crit Critical state

Other Symbols

$\oint_\gamma$ integration around a curve γ

Acronyms / Abbreviations

ALU Arithmetic Logic Unit

BEM Boundary Element Method

CD Contact Dynamics

CFD Computational Fluid Dynamics

CIF Cauchy's Integral Formula

CK Carman - Kozeny

DEM Discrete Element Method

DKT Draft Kiss Tumble

DNS Direct Numerical Simulation

EFG Element-Free Galerkin

FEM Finite Element Method

FLOP Floating Point Operations

FPU Floating Point Unit

FVM Finite Volume Method

GPU Graphics Processing Unit

LBM Lattice Boltzmann Method

LES Large Eddy Simulation

MPM Material Point Method

MRT Multi-Relaxation Time

PCI Peripheral Component Interconnect

PFEM Particle Finite Element Method

PIC Particle-in-cell

PPC Particles per cell

RVE Representative Elemental Volume

SH Savage Hutter

SM Streaming Multiprocessors

USF Update Stress First

USL Update Stress Last

Chapter 1

Introduction

1.1 Motivation and challenge

Combustion has been integral to human development, evolving from early domestic applications such as heating and cooking, through industrial processes including ore smelting, glass production, and ceramic manufacturing, to modern technologies such as steam engines, transportation systems, and power generation [Lackner et al., 2013]. The science of combustion itself has a long and well-established history [Bilger, 2011; Lackner et al., 2013; Allen, 2009; Deane, 1980; Richardson, 2014; Stavrianos, 1999]. Today, the combustion of fossil fuels remains the predominant source of global energy and is expected to retain that role for the foreseeable future owing to the sustained availability of resources such as coal. The BP statistical review of world energy [BP, 2022] reports that fossil fuels accounted for approximately 82% of primary energy consumption as shown in Figure 1.1. Although renewable energy supply increased by over 8 EJ compared with the previous year, overall fossil-fuel consumption was not reduced; primary energy demand rose by 5.8% in 2021 and exceeded 2019 levels by 1.3%, indicating that fossil fuels are likely to continue to dominate the global energy supply. It is well known that combustion not only generates heat that can be converted into power but also produces pollutants such as nitrogen oxides, soot, and unburnt hydrocarbons [Peters, 2000; Lieuwen et al., 2013]. Consequently, the combustion of fossil fuels contributes to environmental degradation and atmospheric pollution through greenhouse gas emissions and toxic species. Increasingly stringent legislations are therefore compelling hard-to-abate sectors, such as aviation, power, and the automotive industry, to reduce pollutant emissions to protect the environment. One approach adopted by these industries is the replacement of conventional hydrocarbon fuels with low or zero carbon alternatives such as hydrogen.

Hydrogen combustion systems are gaining increasing attention as a pathway to meet decarbonisation targets and comply with emerging climate policies. However, hydrogen flames present significant challenges for flame stabilisation, as they are characterised by higher burning velocities, thinner reaction zones, and a greater propensity to propagate close to injector lips while often operating in a diffusion mode [Chaplet et al., 2026]. In addition, hydrogen exhibits a high propensity for

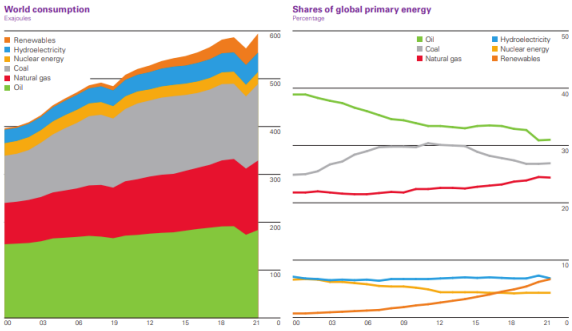

Fig. 1.1 Global primary energy consumption by source (EJ and %) [BP, 2022]

spontaneous ignition compared with conventional fuels, further complicating its safe handling and application [Astbury et al., 2007].Over the past four decades, the progressive refinement of combustion technologies has been a principal factor in lowering atmospheric pollution in industrialised nations [Sawyer, 2009]. In fact, since the mid-1970, combustion science has advanced substantially in response to demands for energy security and environmental protection, driven by progress in mathematical analysis, computational modelling and experimental techniques [Law, 2006]. However, the use of computational approaches is not without its own challenges, as engineers continue to develop methods to address the complexity of reacting flows and turbulence in practical combustion systems, a complexity that requires a detailed interdisciplinary understanding of thermodynamics, chemical kinetics, fluid mechanics, and transport phenomena [Law, 2006].

The continued development of computational fluid dynamics (CFD) has significantly enhanced the role of numerical simulation in modern combustion research for both academic and industrial applications [Poinsot et al., 2005]. Numerical simulation enables detailed three dimensional and time dependent analysis of flame structures with accurate chemical and transport models, supporting the understanding and modelling of combustion phenomena [Mizobuchi, Shinjo, et al., 2005]. It also provides insight into turbulence chemistry interactions by overcoming limitations associated with experimental measurement techniques [Gong et al., 2021], and has therefore been widely employed alongside experimental investigations to study flame stabilisation in turbulent lifted hydrogen jet flames [Brockhinke et al., 2000; Mizobuchi, Shinjo, et al., 2005; Mizobuchi, Tachibana, et al., 2002; Yoo et al., 2009; Cheng et al., 1992; Schumann et al., 2025]. These investigations are typically conducted under lean operating conditions, to improve efficiency, reduce flame temperatures and thereby reduce pollutant emissions [Driscoll, 2011; Massey et al., 2019], although such conditions also increase susceptibility to ignition failure and flame instability [Gicquel et al., 2012].

Combustion frequently involves the interaction of a fuel and an oxidiser, and the way these reactants are mixed determines the type of flame formed, strongly influencing the combustion behaviour [Law, 2006]. The mode of mixing between fuel and oxidiser determines the type of flame formed and consists of premixed combustion in which reactants are initially mixed, non-premixed mode in which mixing occurs through molecular diffusion, and partially premixed combustion

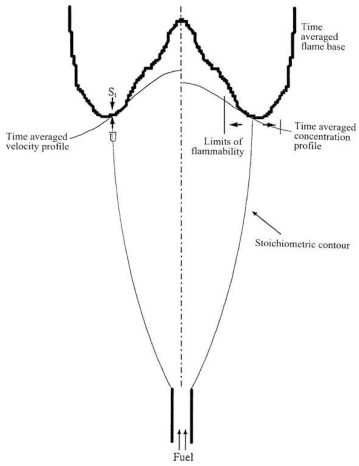

Fig. 1.2 Schematic diagram of the lifted flame [Lawn, 2009]

which arises in practical turbulent systems due to incomplete mixing prior to ignition, making the understanding of flame stabilisation mechanisms essential [Peters, 2000; Law, 2006; Syred, 2006]. Flames in large industrial boilers and gas turbines are commonly stabilised using non premixed jet configurations, with swirling flows employed to extend the range of flame stability in combustion systems; however, as this research does not consider swirl stabilised flows, the reader is referred to the relevant literature for further discussion [Massey et al., 2022; Feikema et al., 1990]. The turbulent lifted hydrogen jet flame arises under conditions of partially premixed combustion and is characterised by a separation between the flame stabilisation root and the burner nozzle, defined as the lift off height H, as shown in Figure 1.2. This configuration involves fuel injected into quiescent air, forming inhomogeneous mixtures where the flame remains attached at low jet velocities but stabilises downstream at higher velocities due to turbulence and local extinction [Peters, 2000], and is commonly preferred in practical systems to protect the burner nozzle [Mansour, 2003]. In a turbulent lifted jet flame, uneven mixing occurs upstream of the flame root, leading to different combustion behaviours at various locations. Hence, combustion is said to be multi-modal [Schumann et al., 2025; Mizobuchi, Shinjo, et al., 2005; Mizobuchi, Tachibana, et al., 2002].

1.2 Approach and objectives

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using ‘Content here, content here’, making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for ‘lorem ipsum’ will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

1.3 Thesis structure

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet..", comes from a line in section 1.10.32.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from “de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham

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Section 1.10.32 of “de Finibus Bonorum et Malorum", written by Cicero in 45 BC: “Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?"

1914 translation by H. Rackham: “But I must explain to you how all this mistaken idea of denouncing pleasure and praising pain was born and I will give you a complete account of the system, and expound the actual teachings of the great explorer of the truth, the master-builder of human happiness. No one rejects, dislikes, or avoids pleasure itself, because it is pleasure, but because those who do not know how to pursue pleasure rationally encounter consequences that are extremely painful. Nor again is there anyone who loves or pursues or desires to obtain pain of itself, because it is pain, but because occasionally circumstances occur in which toil and pain can procure him some great pleasure. To take a trivial example, which of us ever undertakes laborious physical exercise, except to obtain some advantage from it? But who has any right to find fault with a man who chooses to enjoy a pleasure that has no annoying consequences, or one who avoids a pain that produces no resultant pleasure?"

Section 1.10.33 of “de Finibus Bonorum et Malorum", written by Cicero in 45 BC: “At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint occaecati cupiditate non provident, similique sunt in culpa qui officia deserunt mollitia animi, id est laborum et dolorum fuga. Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat."

1914 translation by H. Rackham: “On the other hand, we denounce with righteous indignation and dislike men who are so beguiled and demoralized by the charms of pleasure of the moment, so blinded by desire, that they cannot foresee the pain and trouble that are bound to ensue; and equal blame belongs to those who fail in their duty through weakness of will, which is the same as saying through shrinking from toil and pain. These cases are perfectly simple and easy to distinguish. In a free hour, when our power of choice is untrammelled and when nothing prevents our being able to do what we like best, every pleasure is to be welcomed and every pain avoided. But in certain circumstances and owing to the claims of duty or the obligations of business it will frequently occur that pleasures have to be repudiated and annoyances accepted. The wise man therefore always holds in these matters to this principle of selection: he rejects pleasures to secure other greater pleasures, or else he endures pains to avoid worse pains."

Chapter 2

Research Background

2.1 Governing equations

2.2 Overview of turbulent combustion model

2.2.1 First subsection in the first section

. . . and some moreing

2.2.2 First subsection in the first section

. . . and some more

2.2.3 First subsection in the first section

. . . and some more

2.2.3.1 First subsection in the first section

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2.3 Overview of turbulent combustion model

2.3.1 First subsection in the first section

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2.3.1.1 First subsection in the first section

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2.3.1.2 First subsection in the first section

. . . and some moreing

I'm going to randomly include a picture Figure 2.1.

If you have trouble viewing this document contact Krishna at: kks32@cam.ac.uk or raise an issue at https://github.com/kks32/phd-thesis-template/

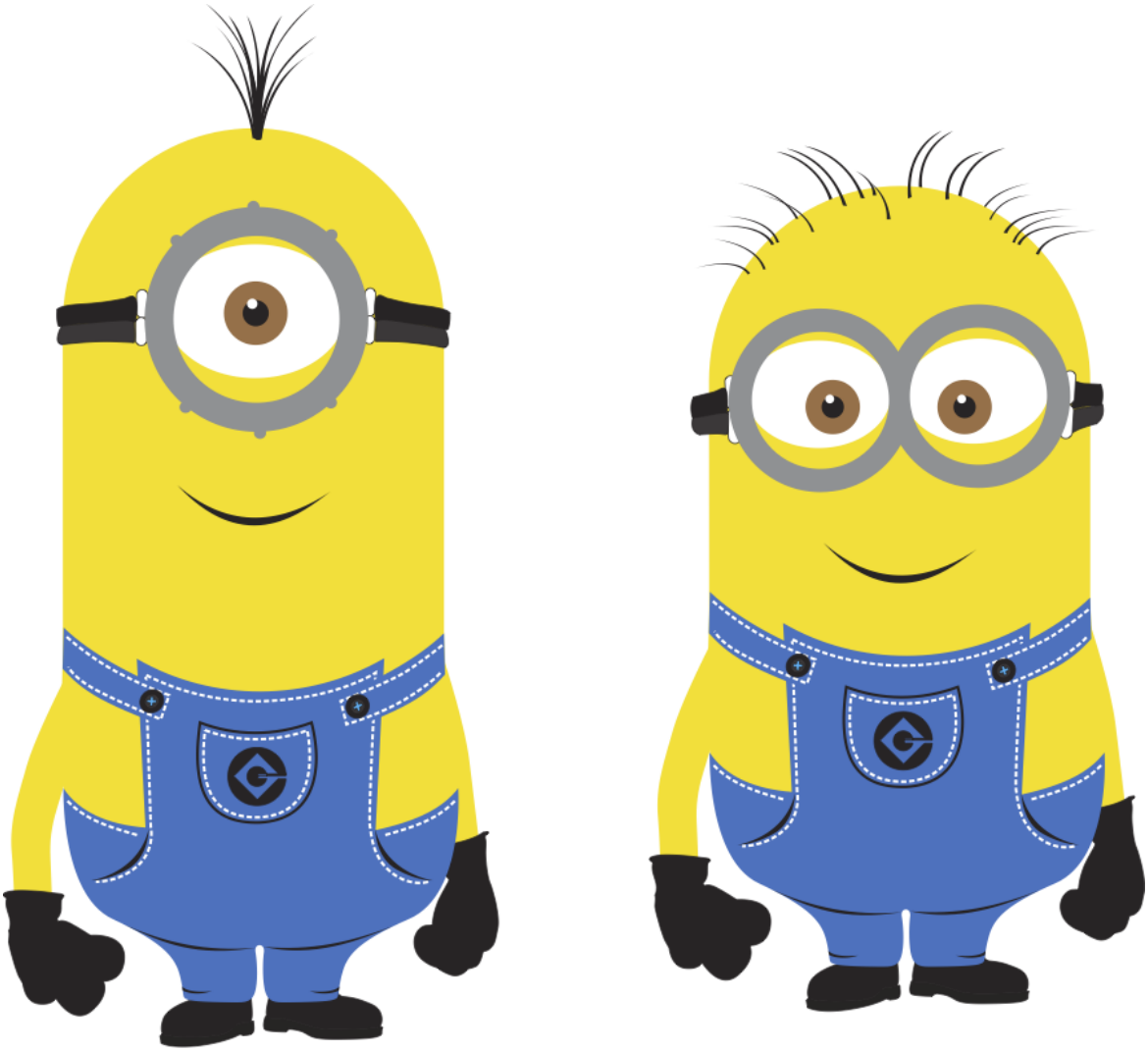

Fig. 2.1 This is just a long figure caption for the minion in Despicable Me from Pixar

Enumeration

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2. The second topic is duller
   (a) The first subtopic is silly
   (b) The second subtopic is stupid
3. The third topic is the dullest

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  - The first subtopic is silly
  - The second subtopic is stupid
- The third topic is the dullest

Description

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The second subtopic is stupid

The third topic is the dullest

2.4 Hidden section

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[1] My footnote goes blah blah blah! ...

(a) Tom and Jerry (b) Wall-E (c) Minions

Fig. 2.2 Best Animations

Subplots

I can cite Wall-E (see Fig. 2.2b) and Minions in despicable me (Fig. 2.2c) or I can cite the whole figure as Fig. 2.2

Chapter 3

My third chapter

3.1 First section of the third chapter

And now I begin my third chapter here . . .

And now to cite some more people [**Rea85**; **Ancey1996**]

3.1.1 First subsection in the first section

. . . and some more

3.1.2 Second subsection in the first section

. . . and some more . . .

3.1.2.1 First subsub section in the second subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it . . .

3.1.3 Third subsection in the first section

. . . and some more . . .

3.1.3.1 First subsub section in the third subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it and some more and some more and some more and some more and some more and some more and some more . . .

3.1.3.2 Second subsub section in the third subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it . . .

3.2 Second section of the third chapter

and here I write more . . .

3.3 The layout of formal tables

This section has been modified from "Publication quality tables in LaTeX*" by Simon Fear.

The layout of a table has been established over centuries of experience and should only be altered in extraordinary circumstances.

When formatting a table, remember two simple guidelines at all times:

1. Never, ever use vertical rules (lines).
2. Never use double rules.

These guidelines may seem extreme but I have never found a good argument in favour of breaking them. For example, if you feel that the information in the left half of a table is so different from that on the right that it needs to be separated by a vertical line, then you should use two tables instead. Not everyone follows the second guideline:

There are three further guidelines worth mentioning here as they are generally not known outside the circle of professional typesetters and subeditors:

3. Put the units in the column heading (not in the body of the table).
4. Always precede a decimal point by a digit; thus 0.1 *not* just .1.
5. Do not use 'ditto' signs or any other such convention to repeat a previous value. In many circumstances a blank will serve just as well. If it won't, then repeat the value.

A frequently seen mistake is to use '\begin{center}' . . . '\end{center}' inside a figure or table environment. This center environment can cause additional vertical space. If you want to avoid that just use '\centering'

Table 3.1 A badly formatted table

	Species I		Species II	
Dental measurement	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

Table 3.2 A nice looking table

Dental measurement	Species I		Species II	
	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

Table 3.3 Even better looking table using booktabs

Dental measurement	Species I		Species II	
	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

Chapter 4

My fourth chapter

4.1 First section of the third chapter

And now I begin my third chapter here . . .

And now to cite some more people [**Rea85**; **Ancey1996**]

4.1.1 First subsection in the first section

. . . and some more

4.1.2 Second subsection in the first section

. . . and some more . . .

4.1.2.1 First subsub section in the second subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it . . .

4.1.3 Third subsection in the first section

. . . and some more . . .

4.1.3.1 First subsub section in the third subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it and some more and some more and some more and some more and some more and some more and some more . . .

4.1.3.2 Second subsub section in the third subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it . . .

4.2 Second section of the third chapter

and here I write more . . .

4.3 The layout of formal tables

This section has been modified from "Publication quality tables in LaTeX*" by Simon Fear.

The layout of a table has been established over centuries of experience and should only be altered in extraordinary circumstances.

When formatting a table, remember two simple guidelines at all times:

1. Never, ever use vertical rules (lines).
2. Never use double rules.

These guidelines may seem extreme but I have never found a good argument in favour of breaking them. For example, if you feel that the information in the left half of a table is so different from that on the right that it needs to be separated by a vertical line, then you should use two tables instead. Not everyone follows the second guideline:

There are three further guidelines worth mentioning here as they are generally not known outside the circle of professional typesetters and subeditors:

3. Put the units in the column heading (not in the body of the table).
4. Always precede a decimal point by a digit; thus 0.1 *not* just .1.
5. Do not use 'ditto' signs or any other such convention to repeat a previous value. In many circumstances a blank will serve just as well. If it won't, then repeat the value.

A frequently seen mistake is to use '\begin{center}' . . . '\end{center}' inside a figure or table environment. This center environment can cause additional vertical space. If you want to avoid that just use '\centering'

Chapter 5

My fourth chapter

5.1 First section of the third chapter

And now I begin my third chapter here . . .

And now to cite some more people [**Rea85**; **Ancey1996**]

5.1.1 First subsection in the first section

. . . and some more

5.1.2 Second subsection in the first section

. . . and some more . . .

5.1.2.1 First subsub section in the second subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it . . .

5.1.3 Third subsection in the first section

. . . and some more . . .

5.1.3.1 First subsub section in the third subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it and some more and some more and some more and some more and some more and some more and some more . . .

5.1.3.2 Second subsub section in the third subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it . . .

5.2 Second section of the third chapter

and here I write more . . .

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Appendix A

How to install LaTeX

Windows OS

TeXLive package - full version

1. Download the TeXLive ISO (2.2GB) from
   https://www.tug.org/texlive/
2. Download WinCDEmu (if you don't have a virtual drive) from
   http://wincdemu.sysprogs.org/download/
3. To install Windows CD Emulator follow the instructions at
   http://wincdemu.sysprogs.org/tutorials/install/
4. Right click the iso and mount it using the WinCDEmu as shown in
   http://wincdemu.sysprogs.org/tutorials/mount/
5. Open your virtual drive and run setup.pl

or

Basic MikTeX - TeX distribution

1. Download Basic-MiKTeX(32bit or 64bit) from
   http://miktex.org/download
2. Run the installer
3. To add a new package go to Start » All Programs » MikTex » Maintenance (Admin) and choose Package Manager
4. Select or search for packages to install

TexStudio - TeX editor

1. Download TexStudio from
   http://texstudio.sourceforge.net/#downloads

2. Run the installer

Mac OS X

MacTeX - TeX distribution

1. Download the file from
   https://www.tug.org/mactex/

2. Extract and double click to run the installer. It does the entire configuration, sit back and relax.

TexStudio - TeX editor

1. Download TexStudio from
   http://texstudio.sourceforge.net/#downloads

2. Extract and Start

Unix/Linux

TeXLive - TeX distribution

Getting the distribution:

1. TexLive can be downloaded from
   http://www.tug.org/texlive/acquire-netinstall.html.

2. TexLive is provided by most operating system you can use (rpm,apt-get or yum) to get TexLive distributions

Installation

1. Mount the ISO file in the mnt directory

   ```
   mount -t iso9660 -o ro,loop,noauto /your/texlive####.iso /mnt
   ```

2. Install wget on your OS (use rpm, apt-get or yum install)

3. Run the installer script install-tl.

```
cd /your/download/directory
./install-tl
```

4. Enter command ‘i’ for installation
5. Post-Installation configuration:
   http://www.tug.org/texlive/doc/texlive-en/texlive-en.html#x1-320003.4.1
6. Set the path for the directory of TexLive binaries in your .bashrc file

For 32bit OS

For Bourne-compatible shells such as bash, and using Intel x86 GNU/Linux and a default directory setup as an example, the file to edit might be

```
edit $~/.bashrc file and add following lines
PATH=/usr/local/texlive/2011/bin/i386-linux:$PATH;
export PATH
MANPATH=/usr/local/texlive/2011/texmf/doc/man:$MANPATH;
export MANPATH
INFOPATH=/usr/local/texlive/2011/texmf/doc/info:$INFOPATH;
export INFOPATH
```

For 64bit OS

```
edit $~/.bashrc file and add following lines
PATH=/usr/local/texlive/2011/bin/x86_64-linux:$PATH;
export PATH
MANPATH=/usr/local/texlive/2011/texmf/doc/man:$MANPATH;
export MANPATH
INFOPATH=/usr/local/texlive/2011/texmf/doc/info:$INFOPATH;
export INFOPATH
```

Fedora/RedHat/CentOS:

```
sudo yum install texlive
sudo yum install psutils
```

SUSE:

```
sudo zypper install texlive
```

How to install LaTeX

Debian/Ubuntu:

```
sudo apt-get install texlive texlive-latex-extra
sudo apt-get install psutils
```

Appendix B

Installing the CUED class file

LaTeX.cls files can be accessed system-wide when they are placed in the <texmf>/tex/latex directory, where <texmf> is the root directory of the user's TeXinstallation. On systems that have a local texmf tree (<texmflocal>), which may be named "texmf-local" or "localtexmf", it may be advisable to install packages in <texmflocal>, rather than <texmf> as the contents of the former, unlike that of the latter, are preserved after the LaTeXsystem is reinstalled and/or upgraded.

It is recommended that the user create a subdirectory <texmf>/tex/latex/CUED for all CUED related LaTeXclass and package files. On some LaTeXsystems, the directory look-up tables will need to be refreshed after making additions or deletions to the system files. For TeXLive systems this is accomplished via executing "texhash" as root. MIKTeXusers can run "initexmf -u" to accomplish the same thing.

Users not willing or able to install the files system-wide can install them in their personal directories, but will then have to provide the path (full or relative) in addition to the filename when referring to them in LaTeX.

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